Master thesis notes

Giacomo Galliani

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1 NLO computations in the subtraction scheme

1.1 The singularities of the NLO real corrections for DY processes

Consider the production of a lepton pair, by annichilation of a quark and a antiquark. These process are known as DY process. At LO (in α_s), there are two diagrams, when the exchange is mediated by a photon or a Z boson. The amplitude is known. We label p_1 the momentum of the quark, and p_2 of the anti-quark. The two diagrams are in figure 1.1, and we write their amplitudes as

$$\mathcal{M}(1,2) = \delta_{ij}\bar{v}(p_2)\gamma^\mu u(p_1)L_\mu(p_\ell, p_{\bar{\ell}}) \tag{1.1}$$

The function L_μ contains the propagator of the photon/Z boson, as well as the lepton vertex. So we just used the electromagnetic vertex of the quarks.

As is well known, if we have to compute the cross section at NLO, there are two type of corrections we need to compute. Naturally there are the 1 loop diagrams, that are called virtual, because in the sub-process, an off-shell virtual particle is created. Then there are the real diagrams: in those, an extra gluon appears in the final space. This processes are real because in those a real on-shell gluon is created, and in particular the final phase space is different. Real diagrams are tree level: they should be easy, but the integration over the final state gluon gives some trouble.

Let us discuss in detail. In figure 1.2 we've drawn the relevant diagrams, both real and virtual. Let's focus on the real. Their amplitude is obtained by eq. 1.1, by adding the multiplicative factor of the extra color vertex, times the quark propagator, that can take values of $p_1 - p_4$, or $-(p_2 - p_4)$, if p_4 is the momentum of the emitted gluon.

Thus, we write

Figura 1.1: the LO diagrams for DY process

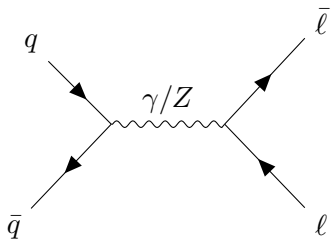


Figura 1.2: NLO diagrams for DY process



$$\mathcal{M}(1,2,4) = \epsilon^*_\nu(p_4) g_{s,b} (T^a)_{ij} \bar{v}(p_2) \left(\gamma^\mu \frac{\not{p}_1 - \not{p}_4}{(p_1 - p_4)^2} \gamma^\nu - \gamma^\nu \frac{\not{p}_2 - \not{p}_4}{(p_2 - p_4)^2} \gamma^\mu \right) u(p_1) L_\mu(p_\ell, p_{\bar{\ell}}) \tag{1.2}$$

1.1.1 The soft limit

To simplify things, we'd like to use Dirac's equation, that tells us $\not{p}_1 u(p_1) = 0 = \bar{v}(p_2)\not{p}_2$ (we will always consider massless quarks). In both cases, we anti-commute with the γ^ν Dirac gamma,

$$\begin{aligned} \not{p}_1 \gamma^\nu u(p_1) &= \left\{ \not{p}_1, \gamma^\nu \right\} u(p_1) = 2p_1^\nu u(p_1) \\ \bar{v}(p_2)\gamma^\nu \not{p}_2 &= 2\bar{v}(p_2) p_2^\nu \end{aligned} \tag{1.3}$$

Now consider the "soft" limit, where $E_4 \to 0$: then we can take $\not{p}_4 = 0$ in the numerator. For the denominator instead we use $(p_1 - p_4)^2 = -2p_1 \cdot p_4 = -2E_1 E_4 (1 - \cos\theta_{14}) = -2E_1 E_4 \rho_{14}$. In general, we indicate $\rho_{ij} = (1 - n_i \cdot n_j)$, if n_i are unit vector that point in the same direction as p_i. Therefore $\rho_{ij} \in [0, 2]$, $\forall i, j$.

We define $S_i A := \lim_{E_i \to 0} A$. We find

$$S_4 \mathcal{M}(1,2,4) = g_{s,b} \epsilon^*_\nu(p_4) (T^a)_{ij} \bar{v}(p_2) \left(-\frac{\gamma^\mu p_1^\nu}{E_1 E_4 \rho_{14}} + \frac{\gamma^\mu p_2^\nu}{E_2 E_4 \rho_{24}} \right) u(p_1) L_\mu(p_\ell, p_{\bar{\ell}}) \tag{1.4}$$

squaring this, we can write

$$\begin{aligned} S_4 |\mathcal{M}(1,2,4)|^2 &= -\frac{g_{s,b}^2 (N_c^2 - 1)}{2N_c E_4^2} \left(-\frac{p_1^\nu}{E_1 \rho_{14}} + \frac{p_2^\nu}{E_2 \rho_{24}} \right)^2 |\mathcal{M}(1,2)|^2 \\ &= 2\frac{g_{s,b}^2 C_F}{E_4^2} \left(\frac{p_1 \cdot p_2}{E_1 E_2 \rho_{14} \rho_{24}} \right) |\mathcal{M}(1,2)|^2 = 2\frac{g_{s,b}^2 C_F}{E_4^2} \left(\frac{\rho_{12}}{\rho_{14}\rho_{24}} \right) |\mathcal{M}(1,2)|^2 \end{aligned} \tag{1.5}$$

In the first line, the factor $N_c^2 - 1$ comes from $\mathrm{Tr}(T^a T^a) = \frac{1}{2}\delta^{aa} = \frac{N_c^2 - 1}{2}$, and the factor $N_c^- 1$ is the color factor of $|\mathcal{M}(1,2)|^2$. Thus we see $C_F = \frac{N_c^2 - 1}{2N_c}$ appearing naturally. This is the known factorization of the matrix element in the soft limit, and the factor before $|\mathcal{M}(1,2)|^2$ is called *eikonal* factor. In can be shown that this is happens universally, not only for DY process.

Now that we have properly obtained the eikonal factor, we see that its integral over the gluon finale phase space is not finite. Of course this is a problem: we cannot, for example, put the integral in a MC program and get a sensibile result.

This is not the only singularity of the amplitude squared. In the soft limit, the denominator cointains the factors $\rho_{14}\rho_{24} \propto (1-\cos\theta_{14})(1-\cos\theta_{24})$ that go to zero as θ^2 as $\theta \to 0$. Thus also the angular integral is divergent.

In this section we focused on a particular process (DY), but we used a rather general approach. Indeed, one can show that the factorization of the amplitude in the soft limit is universal, and given by the following formula (Catani, 2000):

$$\lim_{E_q \to 0} |\mathcal{M}(p_1,\dots,p_n;q)|^2 = -4\pi\alpha_s\mu^{2\epsilon}\sum_{ij} S_{ij}\vec{T}_i\cdot\vec{T}_j|\mathcal{M}(p_1,\dots,p_n)|^2, \quad S_{ij} = \frac{p_i\cdot p_j}{(p_i\cdot q)(p_j\cdot q)} \tag{1.6}$$

Let's show that this reproduces eq 1.5. Remember that $g_{s,b}^2 = 4\pi\alpha_s$. For now, let's not worry about UV divergences and the running of α_s. The new part is computing the color products. In general, one can compute them using $(\vec{T}_q)^2 = (\vec{T}_{\bar{q}})^2 = C_F$, as well as color conservation. The sum runs over the distinct indexes i, j and is symmetric in the exchange. So we find

$$-2g_{s,b}^2\frac{p_i\cdot p_j}{(p_i\cdot q)(p_j\cdot q)}(\vec{T}_q\cdot\vec{T}_{\bar{q}})|\mathcal{M}(1,2)|^2 \tag{1.7}$$

Using color conservation,

$$\vec{T}_q + \vec{T}_{\bar{q}} = 0 \implies (\vec{T}_q + \vec{T}_{\bar{q}})^2 = 2C_F + 2\vec{T}_q\cdot\vec{T}_{\bar{q}} = 0 \implies \vec{T}_q\cdot\vec{T}_{\bar{q}} = -C_F \tag{1.8}$$

So we recover our previous expression.

1.1.2 The collinear limit

That the angular integral is divergent in the collinear limit (i.e. when one of the θs goes to zero) is true also when the gluon emmitted is *hard*. To see this we go back to the amplitude, eq. 1.2.

1.2 Dealing with IR divergences: pdfs renormalization

If the result is infinite, at some points in our computation we must have done a mistake. Actually, the mistakes are two:

1. We did not consider the virtual processes that contribute to the same order in α_s.
2. We computed a partonic cross section: alone, it is not a physical observable as we can't have free partons in the initial state.

To obtain a sensible result, first we need to add up the virtual corrections, and then we need to integrate the cross section using the partonic distribution functions (pdfs). Thus, we can write the pdfs as a sum of a bare pdf and a counterterm, that eliminates the remaining poles.

The *hadronic* (thus observable) cross section is given from $F_{LM}(1,2)$ as

$$\int_0^1 dx_1 dx_2 f_q^{bare}(x_1) f_q^{bare}(x_2)\frac{1}{x_1P_1x_2P_2}F_{LM}(x_1P_1, x_2P_2) \tag{1.9}$$

where P_1, P_2 are the momenta of the incoming hadrons, and we recognize the flux factor $(2s)^{-1} = (x_1 P_1 x_2 P_2)^{-1}$.

As done for the couplings or masses, the counterterms for the pdfs can be computed as a perturbative series in α_s. In the $\bar{\text{MS}}$ scheme, one writes the bare pdfs in terms of the remormalized using

$$f_i^{bare}(x) = [\delta_{ij}\delta(1-y_2) + \frac{\alpha_s(\mu^2)}{2\pi}\frac{1}{\epsilon}\hat{P}_{ij}^{(0)}(y_1) + o(\alpha_s^2)] \otimes f_j^{ren}(y_2) \tag{1.10}$$

where $\otimes$ is the convolution. $\hat{P}_{ij}^{(0)}$ is instead the Altarelli-Parisi kernel function. Inserting this in equation 1.9, considering for now only the terms in x_1, one gets

$$\begin{aligned}
&\int_0^1 dx_1 f_q^{bare}(x_1)\frac{F_{LM}(x_1P_1,\ldots)}{x_1} = \\
&\int_0^1 dx_1 f_q^{ren}(x_1)\frac{F_{LM}(x_1P_1,\ldots)}{x_1} + \int_0^1 dx_1 \frac{\alpha_s(\mu^2)}{2\pi\epsilon}[\hat{P}_{qq}^{(0)} \otimes f_q^{ren}](x_1)\frac{F_{LM}(x_1P_1,\ldots)}{x_1}
\end{aligned} \tag{1.11}$$

We recognize in the second term a counterterm of order α_s. Writing explicitly the convolution, and integrating over x_1 using the delta function, we obtain:

$$\begin{aligned}
&\int_0^1 dx_1 dy_1 dy_2 \frac{\alpha_s(\mu^2)}{2\pi x_1 \epsilon}\hat{P}_{qq}(y_1) f_q^{ren}(y_2) F_{LM}(x_1P_1,\ldots)\delta(x_1 - y_1y_2) = \\
&\int_0^1 dy_1 dy_2 \frac{\alpha_s(\mu^2)}{2\pi y_1 y_2 \epsilon}\hat{P}_{qq}^{(0)}(y_1) f_q^{ren}(y_2) F_{LM}(y_1y_2P_1,\ldots) = \\
&\int_0^1 dx_1 \frac{f_q^{ren}(x_1)}{x_1}\int_0^1 dz \frac{\alpha_s(\mu^2)}{2\pi z}\frac{1}{\epsilon}\hat{P}_{qq}^{(0)}(z) F_{LM}(zx_1P_1,\ldots)
\end{aligned} \tag{1.12}$$

We recognize therefore a *partonic* counterterm, given by

$$d\sigma^{CV,1} = \frac{\alpha_s(\mu^2)}{2\pi\epsilon}\int_0^1 dz \hat{P}_{qq}(z)\frac{F_{LM}(z\cdot 1, 2)}{z} \tag{1.13}$$

Adding the contribution to the other renormalized pdf ($f_q^{bare}(x_2)$), we get the form of the explicit counterterm:

$$d\sigma^{CV} = \frac{\alpha_s(\mu^2)}{2\pi\epsilon}\int_0^1 dz \hat{P}_{qq}^{(0)}(z)\left[\frac{F_{LM}(z\cdot 1, 2)}{z} + \frac{F_{LM}(1, z\cdot 2)}{z}\right] \tag{1.14}$$

1.3 The subtraction scheme for $q\bar{q} \to V + g$

As mentioned before, if an integral is divergent, we cannot integrate it numerically. A possibile strategy for dealing with this problem is defining some operators that extract the singularities. Then one can try to compute analitically the integral in the particular limit, and use a numerical algorithm (e.g. Monte Carlo integrator) to compute the regular integral.

To see how we can do this in practice, first we introduce some useful notation. We also switch to the more general case where the process is $q\bar{q} \to V + g$, with V colorless particles. We denote the contribution of the real diagrams of $d\sigma^{NLO}$ as

$$d\sigma^R = \frac{1}{2s}\int [dg_4] F_{LM}(1,2,4) \tag{1.15}$$

where $(2s)^{-1}$ is the flux factor, $[dg_4] = \frac{d^{d-1}p_4\theta(E_{max}-E_4)}{(2\pi)^d 2E_4}$ and

$$F_{LM}(1,2,4) = d\text{Lips}_V |\mathcal{M}(1,2,4)|^2 \mathcal{F}_{kin}(1,2,4) \tag{1.16}$$

where $d\text{Lips}_V$ is the lorentz invariant phase space of the particles V, and $\mathcal{F}_{kin}(1,2,4)$ is a IR safe observable.

Now notice that the soft limit is not the only point in the phase space that is not integrable. There is also the so called *collinear* singularity, that arises if θ_{14} or θ_{24} vanishes (or, equivalently, ρ_{12} or ρ_{14} go to 0). To be able to integrate the matrix element numerically we must therefor subtract both the soft and the collinear divergences. To this aim we also define the collinear operator,

$$C_{ij}A = \lim_{\rho_{ij}\to 0} A \tag{1.17}$$

The isoleted divergences of $F_{LM}(1,2,4)$ are given by $C_{41}F_{LM}(1,2,4)$, $C_{42}F_{LM}(1,2,4)$, $S_4F_{LM}(1,2,4)$. We thus re-write $d\sigma^R$ as (using the convention where $\int[dg_4]\ldots = <\ldots>$)

$$\begin{aligned} d\sigma^R &= \frac{1}{2s} < (C_{41}+C_{42})F_{LM}(1,2,4) > + \frac{1}{2s} < (1-(C_{41}+C_{42}))F_{LM}(1,2,4) > \\ &= \frac{1}{2s}(< (C_{41}+C_{42})F_{LM}(1,2,4) > + < (I-C_{41}-C_{42})S_4F_{LM}(1,2,4) > + < \hat{O}_{NLO}F_{LM}(1,2,4) >) \end{aligned} \tag{1.18}$$

where we defined $\hat{O}_{NLO} = (I - C41 - C_{42})(I - S_4)$. The term $\propto \hat{O}_{NLO}$ is now free from any singularity by costruction, and can be numerically computed safely. Hence we must compute only the two extra terms.

Claim 1.3.1

The term $< (I - C_{41} - C_{42})S_4F_{LM}(1,2,4) >$ is zero

To see this, we use the expresion for $S_4F_{LM}(1,2,4)$ that we derived earlier (eq. **??**). In the general case, the formula can be written as

$$S_4F_{LM}(1,2,4) = \frac{2g_{s,b}^2 C_F}{E_4^2}\frac{\rho_{12}}{\rho_{14}\rho_{24}}F_{LM}(1,2) \tag{1.19}$$

We have to compute the action of C_{14} as well as C_{24} on this. Of course the limit affects only the fraction $\frac{\rho_{12}}{\rho_{14}\rho_{24}}$. In the CM frame, $\rho_{12} = 2$. Also, if p_4 is allined with p_1, it must be anti-parallel to p_2. We conclude that

$$C_{14}\frac{\rho_{12}}{\rho_{14}\rho_{24}} = \frac{1}{\rho_{14}}, \quad C_{24}\frac{\rho_{12}}{\rho_{14}\rho_{24}} = \frac{1}{\rho_{24}} \implies (C_{14}+C_{24})\frac{\rho_{12}}{\rho_{14}\rho_{24}} = \frac{1}{\rho_{14}} + \frac{1}{\rho_{24}} = \frac{\rho_{14}+\rho_{24}}{\rho_{14}\rho_{24}} \tag{1.20}$$

Again since we're in the CM frame, $\rho_{14} + \rho_{24} = 2 = \rho_{12}$. We conclude that $C_{14} + C_{42}$ leaves $SF_{LM}(1,2,4)$ unchanged, and so $(I - C_{41} - C_{42})S_4 F_{LM}(1,2,4) = 0$.

The only term that we have to deal with is therefore

$$< (C_{41} + C_{42}) F_{LM}(1,2,4) > \tag{1.21}$$

When we derived the collinear limit factorization for the DY process (eq.), we mentioned that it was completely universal. We can safely write than

$$C_{41} F_{LM}(1,2,4) = \frac{g_{s,b}^2}{E_4^2 \rho_{14}} \frac{1-z}{z} P_{qq}(z) F_{LM}(z \cdot 1, 2) \tag{1.22}$$

The splitting function is defined as before, and $z = 1 - E_4/E_1$. We also simplified the denominator using $(p_1 - p_4)^2 = -2p_1 \cdot p_4 = -2E_1 E_4 \rho_{14} = -2E_4^2 \rho_{14} \frac{E_1}{E_4} = -2\frac{E_4^2 \rho_{14}}{1-z}$.

Problems start to arise when we consider this term integrated over the gluon final phase space. We need to evaluate the integral in dimensional regularization, so we can cancel it with the UV divergences of the virutual NLO corrections. We write then

$$[dg_4] = \frac{E_4^{d-3} dE_4 d\cos\theta (1 - \cos^2\theta)^{d/2-2} d\Omega_{d-2}}{2(2\pi)^{d-1}} \tag{1.23}$$

since $E_4 = |p_4|$. Our integrand (eq. 1.22) depends on E_4 through z, and the angular dependence is only in ρ_{14}^{-1}. Our next step is therefore to compute the angular integral, and to write the energy integral in terms of z.

Let's start from the angular integral, that simply reads

$$\int d\Omega_{d-2} d\cos\theta \frac{(1-\cos^2\theta)^2}{1-\cos\theta} = \int d\Omega_{d-2} \int d\cos\theta (1-\cos\theta)^{d/2-3} (1+\cos\theta)^{d/2-2} \tag{1.24}$$

The last integral strongly resembles the euler β integral definition,

$$B(\alpha,\gamma) = \int_0^1 dx x^{\alpha-1}(1-x)^{\gamma-1} = \beta(\alpha,\gamma) = \frac{\Gamma(\alpha)\Gamma(\gamma)}{\Gamma(\alpha+\gamma)} \tag{1.25}$$

So we use the substitution $x = \frac{1}{2}(1 - \cos\theta)$:

$$\begin{aligned} \int d\Omega_{d-2} 2^{d-4} \int_0^1 dx (x)^{d/2-3} (1-x)^{d/2-2} &= \int d\Omega_{d-2} 2^{d-4} \frac{\Gamma(d/2-2)\Gamma(d/2-1)}{\Gamma(d-3)} \\ &= -\frac{1}{\epsilon 4^\epsilon} \frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \int d\Omega_{d-2} \end{aligned} \tag{1.26}$$

We now use $[\alpha_s] = \int d\Omega_{d-2} \frac{g_{s,b}^2 \mu_0^{2\epsilon}}{2(2\pi)^{d-1}}$ (the factor $\mu_0^{2\epsilon}$ is needed for dimensional reasons, since from now on we will take $d = 4 - 2\epsilon$), and the full integral reads

$$\langle C_{41} F_{LM}(1,2,4)\rangle = -\frac{[\alpha_s]}{\epsilon 4^{1+\epsilon}} \frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \int dE_4 E_4^{d-5} \theta(E_{max}-E) \frac{1-z}{z} P_{qq}(z) F_{LM}(z\cdot 1,2) \tag{1.27}$$

Then, we use $z = 1 - E_4/E_1 \implies E_4 = (1-z)E_1,\ dE_4 = -E_1 dz$,

$$\begin{aligned}\langle C_{41} F_{LM}(1,2,4)\rangle &= -\frac{[\alpha_s]}{\epsilon 4^{1+\epsilon}} \frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \int_{z_{\min}}^1 dz E_1 [(1-z)E_1]^{-(1+2\epsilon)} \frac{1-z}{z} P_{qq}(z) F_{LM}(z\cdot 1,2) \\ &= -\frac{[\alpha_s]}{\epsilon} \frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} (2E_1)^{-2\epsilon} \int_{z_{\min}}^1 \frac{dz}{(1-z)^{2\epsilon}} P_{qq}(z) \frac{F_{LM}(z\cdot 1,2)}{z}\end{aligned} \tag{1.28}$$

Now we need to write to write the z integral in such a way that the combination with the virtual corrections is straightforward, i.e. the pole structure is completely explicit. This is still clearly not the case, as in the limit $z \to 1$ the integral is divergent (or at least would be divergent in the limit $\epsilon \to 0$): this is of course the soft limit. We write $F_{LM}(1,2,4)/z = G(z)$, and we rewrite $P_{qq}(z)$ in the following way:

$$\begin{aligned}P_{qq}(z) &= \frac{C_F}{1-z} + \frac{C_F z^2}{1-z} - C_F\epsilon(1-z) = \frac{2C_F}{1-z} + \frac{C_F(z^2-1)}{1-z} - C_F\epsilon(1-z) \\ &= \frac{2C_F}{1-z} + P_{qq}^{reg}(z)\end{aligned} \tag{1.29}$$

In doing so we explicitly separated P_{qq} in a a piece that is singular in the limit $z \to 1$, $\frac{2C_F}{1-z}$, and one piece that is regular $P_{qq}^{rel} = -C_F((z+1) + \epsilon(1-z))$.

Consider then the integral in z: (we can safely extend the integral to the range $[0,1]$ as F_{LM} contains the final state phase space that is zero if there is not enough energy to produce the final state particles, i.e. if $z < z_{\min}$)

$$\begin{aligned}I_z &= \int_0^1 dz \left[\frac{2C_F}{(1-z)^{1+2\epsilon}} + (1-z)^{-2\epsilon} P_{qq}^{reg}(z)\right] G(z) = \\ &= \int_0^1 dz \left[\frac{2C_F(G(z)-G(1))}{(1-z)^{1+2\epsilon}} + (1-z)^{-2\epsilon} P_{qq}^{reg}(z) G(z)\right] + X\end{aligned} \tag{1.30}$$

The last expression holds only if

$$X = \int_0^1 dz \frac{2C_F G(1)}{(1-z)^{1+2\epsilon}} = \frac{2C_F G(1)}{-2\epsilon} = -\frac{C_F G(1)}{\epsilon} \tag{1.31}$$

We notice than that we've taken the divergent part out of the integral. The first fraction is not divergent anymore as also the numerator now goes to zero as $z \to 1$. More explicitly, we write $\frac{G(z)-G(1)}{(1-z)^{1+2\epsilon}}$as a power series:

$$\begin{aligned}\frac{G(z)-G(1)}{(1-z)^{1+2\epsilon}} &= (1-z)^{-2\epsilon}\left(\frac{G(z)-G(1)}{1-z}\right) = e^{-2\epsilon\log(1-z)}\left(\frac{G(z)-G(1)}{1-z}\right) = \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (2\epsilon)^n \log^n(1-z)}{n!}\left(\frac{G(z)-G(1)}{1-z}\right)\end{aligned} \tag{1.32}$$

The first part of the integral then becomes

$$2C_F \sum_{n=0}^{\infty} \frac{(-2\epsilon)^n}{n!} \int_0^1 \frac{\log^n(1-z)}{(1-z)}(G(z)-G(1)) = 2C_F \sum_{n=0}^{\infty} \frac{(-2\epsilon)^n}{n!} \int_0^1 \mathcal{D}_n(z)G(z) \tag{1.33}$$

The factors $\mathcal{D}_n$ are defined by the last equality and are called *plus-distributions*.

In equation 1.28 the z integral is moltiplied by a pole $1/\epsilon$. Therefore we must expand the integral up to $o(\epsilon)$. The whole integral now reads

$$I_z = -\frac{C_F F_{LM}(1,2)}{\epsilon} + \underbrace{2C_F \sum_{n=0}^{\infty} \frac{(-2\epsilon)^n}{n!} \int_0^1 \mathcal{D}_n(z)G(z)}_{=I_z^1} + \underbrace{\int_0^1 dz(1-z)^{-2\epsilon} P_{qq}^{reg}(z)}_{=I_z^2} \tag{1.34}$$

The first term is already explicit, so we just need to expand I_z^1, I_z^2:

$$\begin{aligned} I_z^1 &= 2C_F \left(\int_0^1 dz D_0(z)G(z) - 2\epsilon \int_0^1 dz D_1(z)G(z) \right) \\ I_z^2 &= \int_0^1 (1-z)^{-2\epsilon} P_{qq}^{reg}(z) = -C_F \int_0^1 dz e^{-2\epsilon \log(1-z)}(z+1-\epsilon(1-z)) = \\ &= -C_F \int_0^1 [(z+1) - 2\epsilon \log(1-z)(z+1) + \epsilon(1-z)G(z) \end{aligned} \tag{1.35}$$

Combining everything, we find:

$$I_z = -\frac{C_F F_{LM}(1,2)}{\epsilon} + C_F \int_0^1 dz(2\mathcal{D}_0 - (1+z)) + \epsilon[2\log(1-z)(z+1) - (1-z) - 4\mathcal{D}_1(z)]G(z) + o(\epsilon^2) \tag{1.36}$$

Our result is then

$$\langle C_{41} F_{LM}(1,2,4)\rangle = -\frac{[\alpha_s]}{\epsilon} \frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} s^{-\epsilon} \left[-C_F \left(\frac{1}{\epsilon} + \frac{3}{2} \right) \langle F_{LM}(1,2)\rangle + \int_0^1 \hat{P}_{qq,R}(z) \langle \frac{F(z\cdot 1,2)}{z} \rangle \right] \tag{1.37}$$

where $\hat{P}_{qq,R}$ is a modified splitting function, explicitly given by

$$\begin{aligned} \hat{P}_{qq,R} &= \hat{P}_{qq}^{(0)} + \epsilon \hat{P}_{qq}^{(\epsilon)} + o(\epsilon^2), \\ \hat{P}_{qq}^{(0)} &= C_F \left(2\mathcal{D}_0 - (1+z) + \frac{3}{2}\delta(1-z) \right) \\ \hat{P}_{qq}^{(\epsilon)} &= C_F(2\log(1-z)(z+1) - (1-z) - 4\mathcal{D}_1(z)) \end{aligned} \tag{1.38}$$

the expression for $\hat{P}_{qq}^{(0)}$ matches the LO Altarelli-Parisi splitting kernel. Notice that to obtain eq. 1.37 we just used our result for the z integral (eq. 1.36) into 1.28, adding and subtracting the factor $-3/2$.

Having obtained this expression for the first collinear limit, the second one will be analogous. We can then write the real cross section, in the explicit form that we wanted:

$$\begin{aligned} 2sd\sigma^R = &\langle \hat{O}_{NLO} F_{LM}(1,2,4)\rangle + 2C_F[\alpha_s]\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}s^{-\epsilon}\left[\left(\frac{1}{\epsilon^2}+\frac{3}{2\epsilon}\right)\langle F_{LM}(1,2)\rangle\right] \\ &- \frac{[\alpha_s]}{\epsilon}\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}s^{-\epsilon}\int_0^1 \hat{P}_{qq,R}(z)\left(\langle\frac{F(z\cdot 1,2)}{z}\rangle + \langle\frac{F(1,z\cdot 2)}{z}\rangle\right) \end{aligned} \tag{1.39}$$

We make some comments on this result. The first part is integrable, and can be integrated numerically. The second part is the most divergent ($\propto \epsilon^{-2}$), and multiplies directly the LO factor, $\langle F_{LM}(1,2)\rangle$. Those poles are what we expect to be canceled together with the poles form the virtual diagrams. Lastly, the last term, proportional to ϵ^{-1}, is proportional to the LO amplitude with rescaled energy. Such factors *do not* arise when computing UV divergences, and therefore must be canceled with the collinear counterterms.

Let's now add the UV poles that come from the virtual diagrams. We use Catani's representation of renormalized one loop amplitudes:

$$2sd\sigma^V = -2[\alpha_s]\cos(\epsilon\pi)\left(\frac{C_F}{\epsilon^2}+\frac{3C_F}{2\epsilon}\right)s^{-\epsilon}\langle F_{LM}(1,2)\rangle + \langle F_{LV}^{fin}(1,2)\rangle \tag{1.40}$$

The sum of the two contributes is thus

$$\begin{aligned} 2sd\sigma^{V+R} = &\langle \hat{O}_{NLO} F_{LM}(1,2,4)\rangle + \langle F_{LV}^{fin}(1,2)\rangle + \\ &2[\alpha_s]s^{-\epsilon}\left(\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}-\cos(\pi\epsilon)\right)\left(\frac{C_F}{\epsilon^2}+\frac{3C_F}{2\epsilon}\right)\langle F_{LM}(1,2)\rangle + \\ &-\frac{[\alpha_s]}{\epsilon}\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}s^{-\epsilon}\int_0^1 dz\hat{P}_{qq,R}\left[\frac{F_{LM}(z\cdot 1,2)}{z}+\frac{F_{LM}(1,z\cdot 2)}{z}\right] \end{aligned} \tag{1.41}$$

Expanding the function $\mathcal{S}(1,2) = s^{-\epsilon}\left(\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}-\cos(\pi\epsilon)\right)\left(\frac{C_F}{\epsilon^2}+\frac{3C_F}{2\epsilon}\right)$ up to $o(1)$, one finds that all the poles are calceled, leaving

$$\begin{aligned} 2sd\sigma^{V+R} = &\langle \hat{O}_{NLO} F_{LM}(1,2,4)\rangle + \langle F_{LV}^{fin}(1,2)\rangle + \frac{2\pi^2 C_F}{3}[\alpha_s]s^{-\epsilon}\langle F_{LM}(1,2)\rangle \\ &-\frac{[\alpha_s]}{\epsilon}\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}s^{-\epsilon}\int_0^1 dz\hat{P}_{qq,R}\left[\frac{F_{LM}(z\cdot 1,2)}{z}+\frac{F_{LM}(1,z\cdot 2)}{z}\right] \end{aligned} \tag{1.42}$$

All is left is adding the pdfs counterterm. In the $\bar{\text{MS}}$ scheme, this means adding the factor in eq. 1.14 to the partonic cross section. We therefore find

$$\begin{aligned} 2sd\sigma^{NLO} = &\langle \hat{O}_{NLO} F_{LM}(1,2,4)\rangle + \langle F_{LV}^{fin}(1,2)\rangle + \frac{2\pi^2 C_F}{3}[\alpha_s]s^{-\epsilon}\langle F_{LM}(1,2)\rangle + \\ &\frac{\alpha_s(\mu^2)}{2\pi\epsilon}\left(1-\frac{\mu^{2\epsilon}s^{-\epsilon}e^{\epsilon\gamma_E}\Gamma(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)\int_0^1 dz\hat{P}^0_{qq}(z)\left[\frac{F_{LM}(z\cdot 1,2)}{z}+\frac{F_{LM}(1,z\cdot 2)}{z}\right] \end{aligned}$$

$$- [\alpha_s]\frac{\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}s^{-\epsilon}\int_0^1 dz\hat{P}^{(\epsilon)}_{qq,R}\left[\frac{F_{LM}(z\cdot 1,2)}{z}+\frac{F_{LM}(1,z\cdot 2)}{z}\right] \tag{1.43}$$

Notice that the third line is free from any pole. Expanding the ϵ dependent terms in the second line up to $o(\epsilon)$ we get:

$$\begin{aligned}\left(1-\frac{\mu^{2\epsilon}s^{-\epsilon}\Gamma(1-\epsilon)}{\Gamma(1-2\epsilon)}\right) &= 1-e^{\epsilon(\log(\mu^2/s)+\gamma_E)}\frac{1+\epsilon\gamma_E}{1+2\epsilon\gamma_E}\\ &= 1-(1-\epsilon(\log(\mu^2/s))+\epsilon\gamma_E)(1-\epsilon\gamma_E)+o(\epsilon^2)=\epsilon\log\left(\frac{s}{\mu^2}\right)+o(\epsilon^2)\end{aligned} \tag{1.44}$$

The poles cancel completely! The final result is then

$$\begin{aligned}2sd\sigma^{NLO} = &\langle\hat{O}_{NLO}F_{LM}(1,2,4)\rangle+\langle F^{fin}_{LV}(1,2)\rangle+\frac{2\pi^2C_F}{3}\frac{\alpha_s(\mu^2)}{2\pi}\langle F_{LM}(1,2)\rangle+\\ &\frac{\alpha_s(\mu^2)}{2\pi}\int_0^1 dz\left[\hat{P}^{(0)}_{qq}(z)\log\left(\frac{s}{\mu^2}\right)-\hat{P}^{(\epsilon)}_{qq,R}\right]\left[\frac{F_{LM}(z\cdot 1,2)}{z}+\frac{F_{LM}(1,z\cdot 2)}{z}\right]\end{aligned} \tag{1.45}$$

where we also performed the limit $\epsilon\to 0$.

1.4 A first generalization: the case $q\bar{q}\to X+Ng$

For the hadronic process $pp\to X+N$ jets , the full cross section is given in terms of the partonic cross sections as

$$d\sigma=\sum_{a,b}\int dx_1dx_2f_a(x_1,\mu_F)f_b(x_2,\mu_F)d\hat{\sigma}_{ab}(x_1,x_2,\mu_R,\mu_F;O) \tag{1.46}$$

Where μ_R is the renormalization scale, μ_F the factorization scale. O, instead, is an appropriate observable, that also provides an IR safe definitions of the N jets. This means that the tree level amplitude is safe from IR divergences (this is good news of course, because at tree level we wouldn't be able to cancel those divergences adding up the IR divergences of virtual processes, as well as the pdfs counterterms). For the moment, we're interested in $d\hat{\sigma}^{NLO}_{ab}$, that is given (exactly as the last section) by the sum of 3 distinct terms:

$$d\hat{\sigma}^{NLO}_{ab}=d\hat{\sigma}^R+d\hat{\sigma}^V+d\hat{\sigma}^{pdf} \tag{1.47}$$

the three terms are, in order, the real, virtual and pdf counterterm cross section.

Let's start with the real emission cross section. This process is $1_a+1_b\to X+(N+1)g$. It is again a tree level amplitude, with an extra gluon in the final phase space.

An important point is that since the observable O makes sure that at least N partons are resolved, only one gluon can become soft or collinear. We must therefore be able to identify this unresolved parton. We introduce than the damping factors $\Delta^{(i)}$, $i\in 3,\dots,N_p+1$, (N_p is the number of partons involved in the process, so for $pp\to X+Ng$, $N_p=N+2$ and we always leave the indexes $1,2$ for the initial state partons) that are defined in

such way that they go to zero for every i, except if the parton i is unresolved. They also provide a partition of unity,

$$\sum_{i=3}^{N_p+1} \Delta^{(i)} = 1 \tag{1.48}$$

Following the same scheme of the previous section (and using the same notation), we want to compute

$$\langle F_{LM}(1,2;3,\ldots,N_p+1,X)\rangle = \sum_{i=3}^{N_p+1} \langle \Delta^{(i)} F_{LM}(1,2;3,\ldots,N_p+1,X)\rangle \tag{1.49}$$

In this last equation, $F_{LM}(\ldots)$ is completely symmetric in any exchange of the $N+1$ external gluons. Then in each term of the sum, we can re-label the index, in such a way that all the damping functions are identical, and we obtain

$$\sum_{i=3}^{N_p+1} \langle \Delta^{(i)} F_{LM}(1,2;3,\ldots,N_p+1,X)\rangle = \langle (N_p-1)\Delta^{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle \tag{1.50}$$

The factor $N_p - 1$ multiplies the $1/(N_p-1)!$, turning it into $1/(N_p-2)! = 1/N!$, where N are the number of resolved jets. This is therefore the same symmetry factor that sits in the F_{LM} of the virtual processes. To the sake of IR poles combinations we can omit in both functions the explicit symmetry factor.

As we did in the previous section, we now want to extract the divergences, using the operators S_i, C_{ij}. First, we list the action of those operators on the damping functions,

$$S_{\mathfrak{m}}\Delta^{(\mathfrak{m})} = 1, \quad C_{a\mathfrak{m}}\Delta^{(\mathfrak{m})} = 1, \quad C_{i\mathfrak{m}}\delta^{(\mathfrak{m})} = \frac{E_{\mathfrak{m}}}{E_{\mathfrak{m}}+E_i} =: z_{\mathfrak{m},i} \tag{1.51}$$

where $a = 1,2$ and $i = 3,\ldots,N_p+1$, as before. We start by subtracting the soft,

$$\langle F_{LM}(\mathfrak{m})\rangle = \langle S_{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle + \langle \bar{S}_{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle \tag{1.52}$$

For any operator A, we denote $\bar{A} = I - A$. In the previous section, we only had to consider the collinear limit, because in that case we found that acting with the collinear operators on the soft limit leaves the expression invariant. This is not the case anymore, as the gluon can go collinear to both initial state and final state resolved partons.

Therefore we must already write $\langle S_{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle$, expliciting the ϵ pole structure. We use the universal form of the soft gluon limit (eq. 1.6), and we integrate it over the gluon final phase space:

$$\langle S_{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle = -\frac{g_{s,b}^2 \mu^{2\epsilon}}{2(2\pi)^{d-1}} \int dE_{\mathfrak{m}} E_{\mathfrak{m}}^{d-5} \theta(E_{max}-E) d^{d-2}\Omega \sum_{(ij)} \frac{\rho_{ij}}{\rho_{i\mathfrak{m}}\rho_{j\mathfrak{m}}} (\vec{T}_i \cdot \vec{T}_j) \langle F_{LM}\rangle \tag{1.53}$$

From now one we use $d = 4 - 2\epsilon$, and we inserted the scale $\mu^{2\epsilon}$. The energy integral is straightforward,

$$\langle S_{\mathfrak{m}} F_{LM}(\mathfrak{m})\rangle = -\frac{g_{s,b}^2}{2(2\pi)^{d-1}}\left(\frac{E_{\max}}{\mu}\right)^{-2\epsilon}\int d^{d-2}\Omega \sum_{(ij)}\frac{\rho_{ij}}{\rho_{i\mathfrak{m}}\rho_{j\mathfrak{m}}}(\vec{T}_i\cdot\vec{T}_j)\,\langle F_{LM}\rangle \tag{1.54}$$